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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Electric Vehicle Technology) (CBCS) Degree Examinations

May - June 2024

22EVT202 – EV MOTOR DRIVES AND CONTROL

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Draw the equivalent circuit of a separately wound DC motor and obtain the expression for armature voltage, electrical power and mechanical power.	6	L*3	1	3, 5
b)	Consider a DC motor with armature voltage 125V and armature resistance $R_a = 0.4\Omega$. It is running at 1800 rpm under no load condition. i. Calculate the back emf constant K_b ii. When the rated armature current is 30A, calculate the rated torque. iii. Calculate the rated speed.	8	L3	1	3, 5
c)	Describe the four-quadrant operation of DC motor.	6	L2	1	3, 5
2. a)	Consider a DC motor with $R_a = 0.5$, $K_t = 0.8 \text{ Nm/A}$, and $K_b = 0.8 \text{ Vs/rad}$. i. Assume that the motor terminal voltage reaches $V_a^{\max} = 120 \text{ V}$ when the motor runs at a rated speed, $\omega_r = 140 \text{ rad/s}$. Calculate the rated current and rated torque. ii. Assume that a load torque, $T_L = 6 \text{ Nm}$, is applied and that the field is weakened to a half, i.e., $K_t = 0.4 \text{ Vs/rad}$. Determine the speed.	8	L3	1	3, 5
b)	Compare BLDC and PMSM motors in EV aspects.	6	L2	1	3, 5
c)	Describe the EMF and Torque generation phenomenon in DC motors.	6	L2	1	3, 5
Unit – II					
3. a)	Derive the transfer function for the current control loop and speed control loop of the DC motor. Depict the same using the block diagram.	10	L3	2	3, 5
b)	Explain the significance of gain and phase margin with respect to bode plot of the system.	10	L2	2	3, 5
4. a)	Draw the block diagram of PI controller with reference model and obtain the closed loop transfer function.	8	L3	2	3, 5
b)	Describe the load torque observer with a neat block diagram.	6	L2	2	3, 5
c)	Describe the simplified modelling of practical current loop with a neat block diagram.	6	L2	2	3, 5
Unit – III					
5. a)	With induction motor equivalent circuit diagram, derive the equation for air gap power and explain the same.	6	L3	3	3, 5
b)	Describe the space vector modulation (SVM). Mention the steps involved in SVM generation.	8	L3	3	3, 5
c)	List the difference between IM and 3-phase transformer.	6	L2	3	3, 5
6. a)	Describe the operation of stator field-oriented control scheme of IM.	10	L3	3	3, 5
b)	Explain VVVF control of IM along with speed-torque characteristics curve.	10	L3	3	3, 5
Unit – IV					
7. a)	Describe the control block diagram of PMSM using the coordinate transformation map.	8	L2	4	3, 5
b)	Describe the principle of operation of Maximum Torque per Ampere Control in PMSM.	6	L2	4	3, 5
c)	List the types of PMSM along with cross section diagram.	6	L2	4	3, 5

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| 8. | a) | Compare the parameters of SPMSM and IPMSM | 6 | L2 | 4 | 3, 5 |
| | b) | Distinguish how the machine sizing varies under same power based on maximum torque and speed. | 6 | L2 | 4 | 3, 5 |
| | c) | Describe position sensing and control in case of BLDC motor. | 8 | L2 | 4 | 3, 5 |

Unit – V

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| 9. | a) | Comment on need of position and current sensing in an EV motor control? List few of the position sensing apparatus. | 6 | L2 | 5 | 3, 5 |
| | b) | Write a note on resolver and R/D converter. | 6 | L2 | 5 | 3, 5 |
| | c) | Give an overview of PMSM sensor less control technique. | 8 | L2 | 5 | 3, 5 |
| 10. | a) | Write a note on Current Model Estimator for IM. | 8 | L2 | 5 | 3, 5 |
| | b) | Elaborate basic working of Encoders and Hall Position Sensors. | 6 | L2 | 5 | 3, 5 |
| | c) | Current sampling time in motor control inverters is crucial for controller operation. Justify | 6 | L2 | 5 | 3, 5 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
